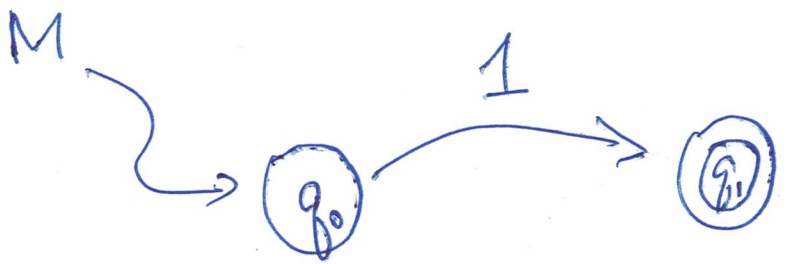
Cpts 317. 9/26

HW Extension? Wed. night.

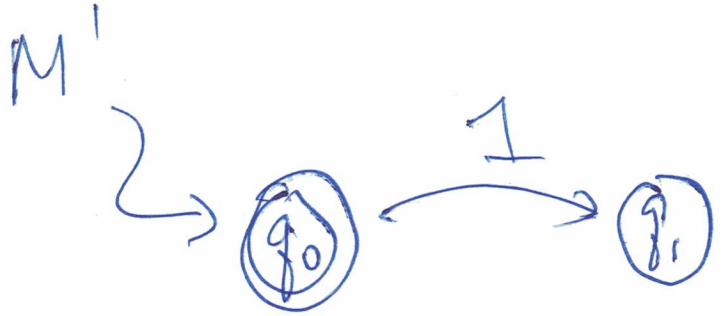
NO. 3. Learn from an example.



$\Longrightarrow$ get M' s.t. $L(M') = \Sigma^* - L(M)$.

$\forall w$,

M accepts w iff M' rejects w.



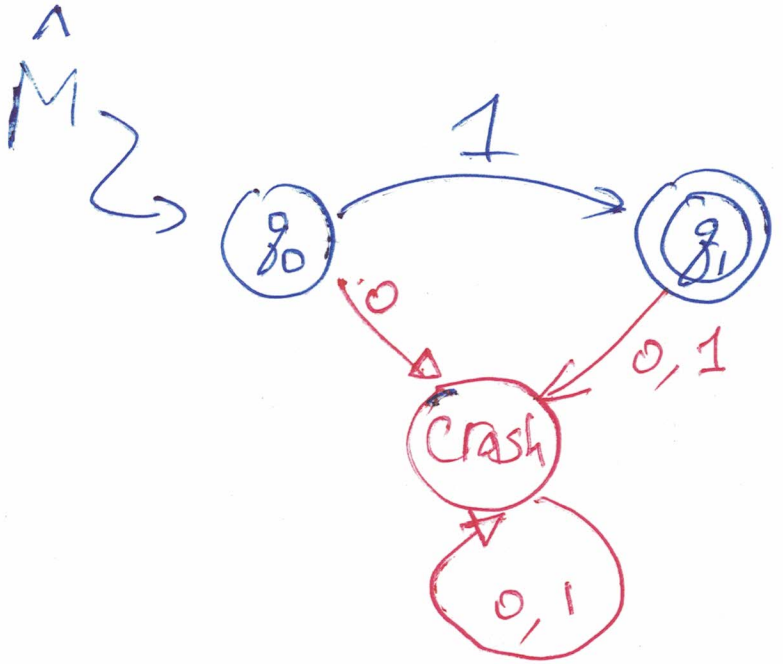
Won't work.

$1 \in L(M)$, $1 \notin L(M')$

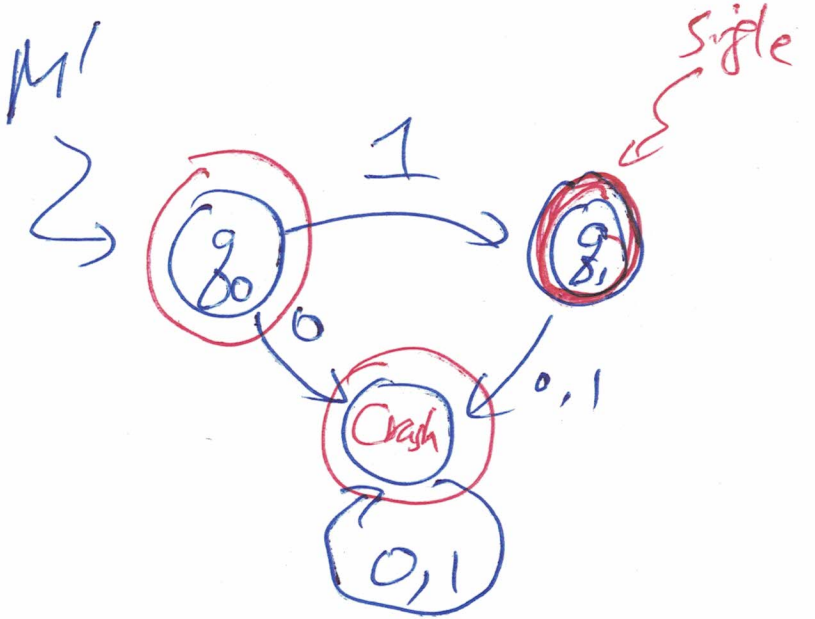
$0 \notin L(M)$, $0 \notin L(M')$.

Correction: make M "complete".

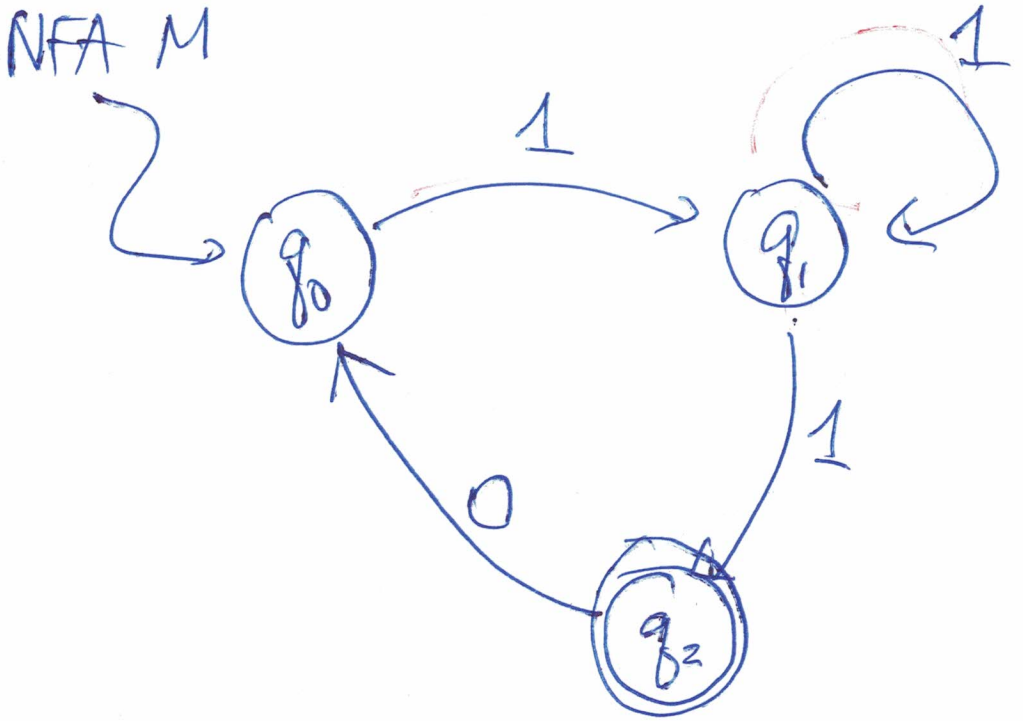
Won't crash!



$\Rightarrow$



NFA M



$11 \in L(M)$, witnessed by $q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_2$ ✓

$01 \notin L(M)$.

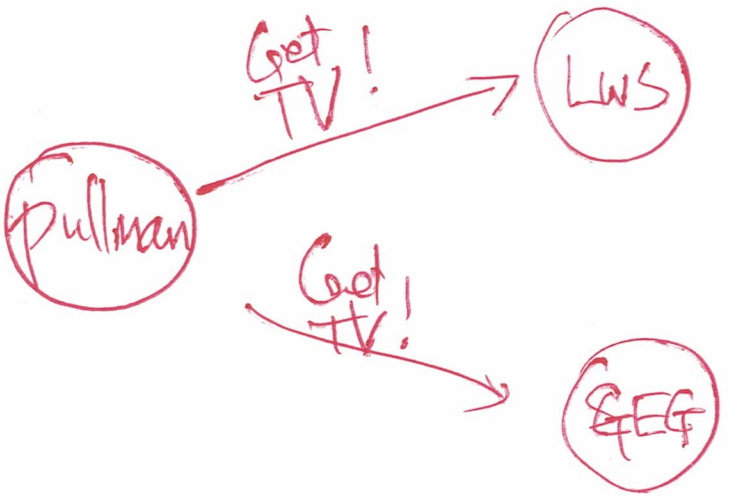
We say that an NFA M accepts w, written $w \in L(M)$, if there is a walk from the initial to accepting state in M on which the input symbols are exactly the w.

// M accepts w $\Rightarrow$ M says yes on w;

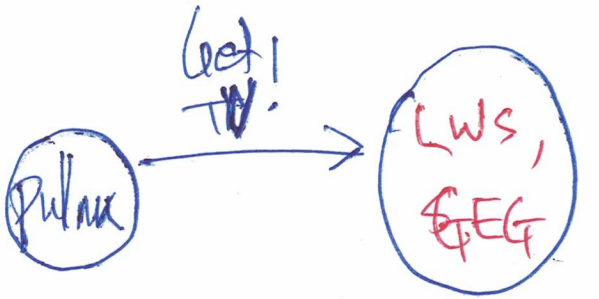
// Can M say NO on w?

Theorem. We have an alg, called subset construction, to convert NFA to DFA (so they accept the same language).

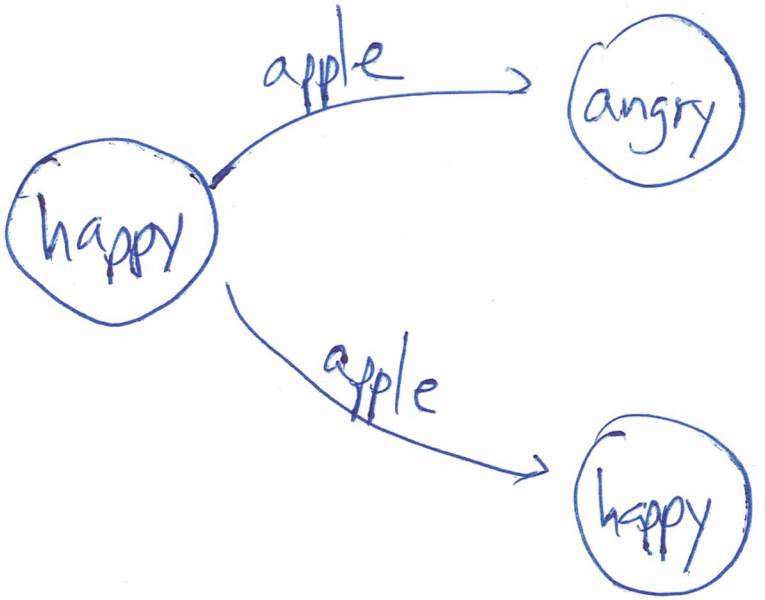
Proof. intuition:



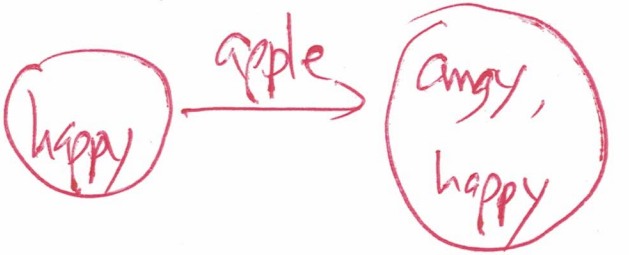
nondet. choice.



no nondet. choice

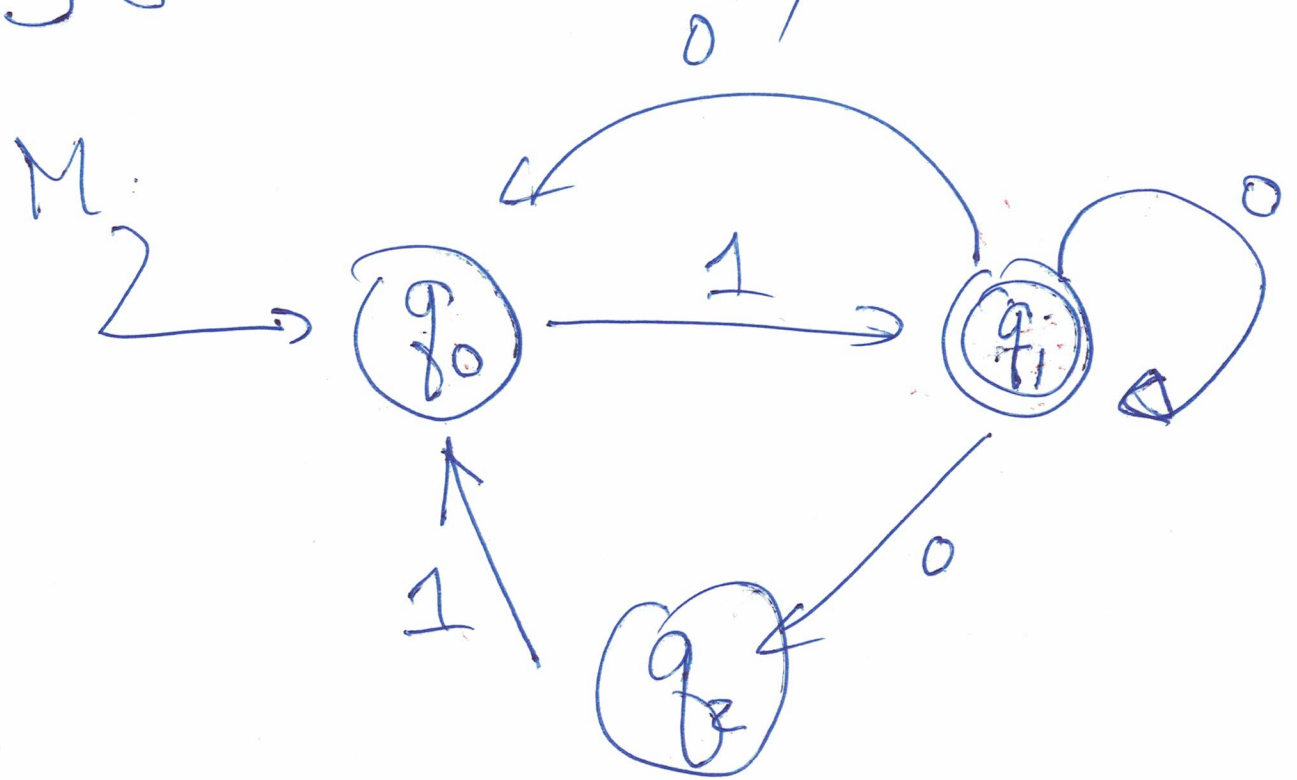


nondet.



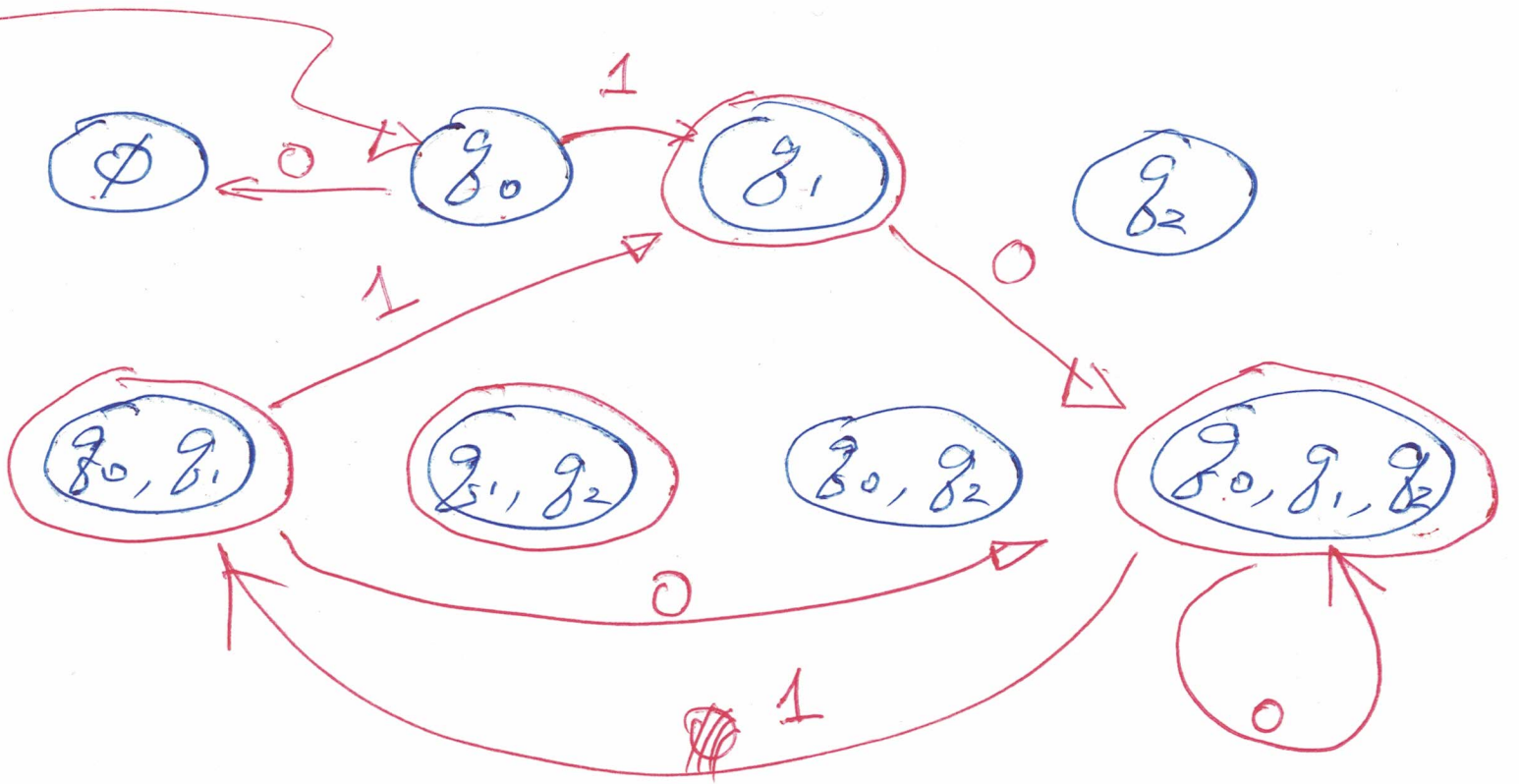
det.

Alg of subset construction by Example:



Want to convert M into DFA:

M'



~~L(M') = 1~~